In [ ]:
```
import cv2
import matplotlib.pyplot as plt
import numpy as np
```

In [ ]:
```
image_normal_path = '/content/Screenshot 2026-08-23 113346.png'
image_underexposed_path = '/content/Screenshot 2026-08-23 113356.png'
image_overexposed_path = '/content/Screenshot 2026-08-23 113406.png'

image_normal = cv2.imread(image_normal_path, cv2.IMREAD_GRAYSCALE)
image_underexposed = cv2.imread(image_underexposed_path, cv2.IMREAD_GRAYSCALE)
image_overexposed = cv2.imread(image_overexposed_path, cv2.IMREAD_GRAYSCALE)

if image_normal is None:
    print(f"Error: Normal image not loaded from {image_normal_path}")
if image_underexposed is None:
    print(f"Error: Underexposed image not loaded from {image_underexposed_path}"
if image_overexposed is None:
    print(f"Error: Overexposed image not loaded from {image_overexposed_path}")
```

In [ ]:
```
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(image_normal, cmap='gray')
plt.title('Normal Image')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(image_underexposed, cmap='gray')
plt.title('Underexposed Image')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(image_overexposed, cmap='gray')
plt.title('Overexposed Image')
plt.axis('off')

plt.show()
```



In [ ]:
```
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.hist(image_normal.ravel(), 256, [0,256])
plt.title('Histogram - Normal Image')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')

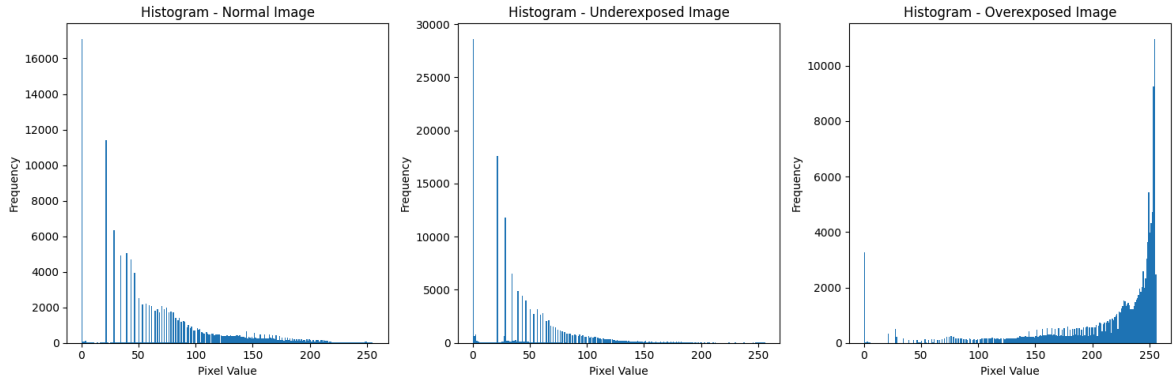
plt.subplot(1, 3, 2)
plt.hist(image_underexposed.ravel(), 256, [0,256])
plt.title('Histogram - Underexposed Image')
plt.xlabel('Pixel Value')
```

```
plt.ylabel('Frequency')

plt.subplot(1, 3, 3)
plt.hist(image_overexposed.ravel(), 256, [0,256])
plt.title('Histogram - Overexposed Image')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_2403/2360260219.py:4: MatplotlibDeprecationWarning: Passing the ra
nge parameter of hist() positionally is deprecated since Matplotlib 3.9; the para
meter will become keyword-only in 3.11.
  plt.hist(image_normal.ravel(), 256, [0,256])
/tmp/ipykernel_2403/2360260219.py:10: MatplotlibDeprecationWarning: Passing the r
ange parameter of hist() positionally is deprecated since Matplotlib 3.9; the par
ameter will become keyword-only in 3.11.
  plt.hist(image_underexposed.ravel(), 256, [0,256])
/tmp/ipykernel_2403/2360260219.py:16: MatplotlibDeprecationWarning: Passing the r
ange parameter of hist() positionally is deprecated since Matplotlib 3.9; the par
ameter will become keyword-only in 3.11.
  plt.hist(image_overexposed.ravel(), 256, [0,256])
```



In [ ]: